

Sports & Nutrition: Key Concepts

This cheatsheet summarizes essential concepts in sports and nutrition, covering balanced diets, macro/micronutrients, weight control, and common food myths.

Balanced Diet

A balanced diet involves consuming the correct proportions of food from all essential food groups. It includes proteins, carbohydrates, fats, vitamins, minerals, and water in appropriate amounts to maintain health.

Nutrition

Nutrition is the dynamic process of consuming food to sustain health and nourish the body.

Macronutrients

These are nutrients required in large amounts, providing energy and building blocks for the body.

Carbohydrates

- Description: Primary energy source, organic compounds of carbon, hydrogen, and oxygen. Crucial for digestive operations.
- Simple Carbohydrates: Soluble in water, crystalline, sweet.
 - Examples: Glucose, fructose, sucrose, lactose.
- Complex Carbohydrates: Not water-soluble, not sweet, non-crystalline.
 - Examples: Starch, glycogen, cellulose.

Proteins

- Description: Contain carbon, oxygen, hydrogen, nitrogen, and sometimes sulfur. Large molecules broken down into amino acids for absorption.
- Requirement: Approximately 0.36 grams of protein per pound of ideal body weight.

Fats

- Description: Composed of carbon, oxygen, and hydrogen. Provide warmth, protect organs, and aid hormone production.
- Saturated Fats: Increase cholesterol levels.
- Unsaturated Fats: Lower cholesterol levels.

Water

- Description: Made of hydrogen and oxygen. Essential for nutrient transportation, waste excretion, body temperature regulation, and various chemical reactions.

Micronutrients

These are nutrients required in small amounts, vital for various bodily functions.

Minerals

- Definition: Chemical elements required as essential nutrients for life-sustaining functions.
- Macro Minerals: Needed in larger quantities.
 - Examples: Calcium, potassium, sodium, magnesium, phosphorus.
- Micro Minerals (Trace Minerals): Needed in smaller quantities.
 - Examples: Iodine, iron, chromium, copper, cobalt.

Vitamins

- Definition: Organic compounds required in minimal amounts to maintain body health.
- Fat-Soluble Vitamins: Stored in the body's fatty tissues and liver.
 - Examples: Vitamin A, Vitamin D, Vitamin E, Vitamin K.
- Water-Soluble Vitamins: Not stored in the body and need to be replenished regularly.
 - Examples: Vitamin B complex (B1, B2, B3, B5, B6, B12, B9, B7), Vitamin C.

Nutritive & Non-Nutritive Components of Diet

Nutritive Components

These components provide energy and essential nutrients.

- Carbohydrates: Primary energy source.
- Proteins: Body building and repair.
- Fats: Energy storage, organ protection.
- Vitamins: Regulate body processes.
- Minerals: Bone health, nerve function, etc.

Non-Nutritive Components

These components do not provide energy but are important for digestion and overall health.

- Fiber: Aids digestion, prevents constipation.
- Water: Hydration, transport.
- Color Compounds: Natural pigments in foods.
- Flavor Compounds: Contribute to taste.
- Plant Compounds: Phytochemicals with health benefits.

Eating For Weight Control

Healthy Weight

- BMI Range: A healthy weight corresponds to a Body Mass Index (BMI) between 19 and 25.
- Overweight: BMI between 25 and 29.
- Obese: BMI of 30 or greater.

Methods to Control Healthy Body Weight

- Set appropriate weight loss goals.
- Reduce calorie intake and maintain an active lifestyle.
- Engage in regular exercise (e.g., gym, yoga).
- Avoid fatty foods, junk foods, and fast foods.
- Do not skip meals; eat smaller, frequent meals.
- Balance calorie intake with calorie expenditure.

The Pitfalls of Dieting

- Diets severely restricting calories (starvation response) slow metabolic rate.
- Loss of muscle tissue lowers metabolic rate.
- Common Pitfalls:
 - Extreme calorie reduction and skipping meals.
 - Restriction of entire food groups (e.g., carbohydrates, fats).
 - Excessive calorie intake through beverages.
 - Lack of regular exercise during dieting.

Food Intolerance

- Definition: Inability of the digestive system to properly process and absorb certain food elements.
- Causes: Absence or inactivity of enzymes needed to break down food.
- Symptoms: Stomach pain, vomiting, headache, nervousness.

Food Myths

Common misconceptions about food:

1. "Potato makes you fat." (Myth: Potatoes are healthy in moderation, preparation matters).
2. "Fat-free products help in losing weight." (Myth: Often high in sugar, not necessarily healthier).
3. "Eggs increase cholesterol level." (Myth: Dietary cholesterol has less impact on blood cholesterol for most people than saturated/trans fats).
4. "Drinking while eating makes you fat." (Myth: Fluids don't add fat; may aid digestion).
5. "Don't take milk immediately after eating fish." (Myth: No scientific basis for this causing harm).